Informal 2. Want $S \subset \mathbb{N}$ with $S \neq \mathbb{N}$ $S \subsetneq \mathbb{N}$ but S is closed under successors. $\rightsquigarrow$ If $n \in S$, then $\sigma(n) \in S$.

Axiom 4 tells us S can't have 0.

$S = \{1, 2, 3, 4, \ldots\} = \mathbb{N}^+$ positive naturals.

$S = \{n \in \mathbb{N} : n \geq 2\}$

$S = \{n \in \mathbb{N} : n \text{ has a nontrivial prime factorization}\}$
$= \{n \in \mathbb{N} : n \geq 2\}$

All have form $\{n \in \mathbb{N} : n \geq k\}$ for some $k \in \mathbb{N}^+$

Def. If $a, b \in \mathbb{N}$ are such that $\sigma(a) = b$, then we say <u>b has predecessor a</u> or <u>a is the predecessor of b</u>.

Theorem 1. The natural number 0 has no predecessor in $\mathbb{N}$.

<u>Proof</u>. For a contradiction, suppose that 0 has a predecessor $x \in \mathbb{N}$. By def. of predecessor $\sigma(x) = 0$. However, this contradicts axiom 2. $\times$ $\Rightarrow\Leftarrow$ $\lightning$

Thus 0 has no predecessor in $\mathbb{N}$. $\square$

Theorem 2: Every nonzero element of $\mathbb{N}$ has a predecessor in $\mathbb{N}$.

Proof. We proceed via Axiom 4. Let S be the set of natural numbers with a predecessor union $\{0\}$. I.e., $S = \{n \in \mathbb{N} : n \text{ has a predecessor}\} \cup \{0\}$. Clearly S contains 0 and is a subset of $\mathbb{N}$. Now suppose $n \in S$. We want to show $\sigma(n)$ is in S. Thus we need to show $\sigma(n)$ has a predecessor. We need to show there is some $k \in \mathbb{N}$ such that $\sigma(k) = \sigma(n)$. But of course n is the predecessor of $\sigma(n)$. Thus $\sigma(n)$ has a predecessor in $\mathbb{N}$, and $\sigma(n) \in S$. We conclude by Axiom 4 that $S = \mathbb{N}$. $\square$

Def. A <u>positive natural number</u> is a nonzero element of $\mathbb{N}$. We denote the positive natural numbers by $\mathbb{N}^+$. In other words, $\mathbb{N}^+ \overset{\text{def}}{:=} \mathbb{N} - \{0\}$

$\mathbb{N} \setminus \{0\}$

Defined to equal.

Ex. Cor. 4. If $n \in \mathbb{N}^+$, then n has a unique predecessor in $\mathbb{N}$.

<u>Proof.</u> By Thm. 2 n has a predecessor in $\mathbb{N}$. To see uniqueness suppose $\sigma(a) = \sigma(b) = n$ with $a, b \in \mathbb{N}$. By Axiom 3, σ is injective. Thus $\sigma(a) = \sigma(b) \Rightarrow a = b$. $\square$

Define a predecessor function $\pi : \mathbb{N}^+ \to \mathbb{N}$ as follows. If $n \in \mathbb{N}^+$, Cor. 4 shows there is a unique $a \in \mathbb{N}$ such that $\sigma(a) = n$. We define $\pi(n)$ to be a. I.e., $\pi(n) \overset{\text{def}}{=} a$ where $\sigma(a) = n$.

Homework define a restricted successor function σ' and show σ' & π are inverses!

Axiom 5 (Iteration): Let $f : S \to S$ be a function. The nth iteration of f is a function from S to itself. $f^n = \underbrace{f \circ f \circ \cdots \circ f}_{n\text{-times}}$. In symbols, $f^n : S \to S$

we don't know what this means.

Such functions satisfy the following

(i) f^0 is the identity function on S.

(ii) $f^{\sigma(n)} = f \circ f^n$.

Exercise 12. Prove that $f^1 = f$.

Idea: $f^1 = f^{\sigma(0)} = f \circ f^0 = f$.

$1 \overset{\text{def}}{=} \sigma(0)$ Axiom 5 (ii) Axiom 5 (i)

Prove $f^2 = f \circ f$

$2 = \sigma(1) \rightarrow$

$$f^{\sigma(1)} = f \circ f^1 = f \circ f.$$

Axiom 5 (ii)

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§ Addition

Def. Let $m, n \in \mathbb{N}$. Recall $\sigma^n : \mathbb{N} \to \mathbb{N}$ is the nth iteration of the successor map. Define $m+n$ to be $\sigma^n(m)$. I.e., $m+n \overset{\text{def}}{=} \sigma^n(m)$

Intuition (flashback) $\boxed{2+3 \overset{\text{def}}{=} \sigma^3(2) = 2+1+1+1 = 5}$

We see that $+$ defines a function from $\mathbb{N}\times\mathbb{N} \longrightarrow \mathbb{N}$. Such functions are called <u>binary operations</u>.

Theorem 4. For all $m \in \mathbb{N}$, one has $m+0 = m$.

Proof: We compute $m+0 = \sigma^0(m) = m$, $\square$

where the first equality is by definition and the second is by Axiom 5 (i).

Lemma 5. $\forall\, m,n \in \mathbb{N}$, $m+\sigma(n) = \sigma(m+n)$

Thm. 6. $\forall\, m \in \mathbb{N}$, $m+1 = \sigma(m)$.

Thm. 7 (Associative Law): For all $x,y,z \in \mathbb{N}$,

$$x+(y+z) = (x+y)+z.$$

<u>Proof</u>. We proceed by induction. Let $x,y \in \mathbb{N}$ be arbitrary. Define $S_{x,y} = \{z \in \mathbb{N} : x+(y+z) = (x+y)+z\}$

First, $S_{x,y} \subset \mathbb{N}$.

We compute $x+(y+0) = x+(y)$ [By Thm. 4]

$= x+y$ [y is alone]

$= (x+y)$ [considering $x+y$]

$= (x+y)+0$ [By Thm. 4].

We see $0 \in S_{x,y}$. Now suppose $n \in S_{x,y}$.

We have $x+(y+\sigma(n)) = x+\sigma(y+n)$ [by Lemma 5]

$= \sigma(x+(y+n))$ [by Lemma 5]

$= \sigma((x+y)+n)$ [since $n \in S_{x,y}$] induction hyp.

$= (x+y)+\sigma(n)$ [by Lemma 5].

We see $\sigma(n) \in S_{x,y}$. Thus Axiom 4 (Induction) shows $S_{x,y} = \mathbb{N}$. Since x,y were arbitrary, $x+(y+z) = (x+y)+z \ \forall x,y,z \in \mathbb{N}$.

Lemma 8. If $n \in \mathbb{N}$, then $\sigma^n(0) = n$.
In particular, $0 + n = n$.

Lemma 9. If $n \in \mathbb{N}$, then $1 + n = \sigma(n)$.

Theorem 10. (Commutativity of $+$)
If $x, y \in \mathbb{N}$, then $x + y = y + x$.

Proof sketch. Use Induction (Axiom 4).

Let $S_x = \{n \in \mathbb{N} : x + n = n + x\}$.

Lemma 8 gives the base case: $x + 0 \overset{\text{Thm. 4}}{=} x \overset{\text{Lemma 8.}}{=} 0 + x$ $\Rightarrow 0 \in S_x$

Suppose $j \in S_x$.

show $x + \sigma(j) = \sigma(j) + x$ using Lemma 9 & induction hyp.

§1.5: Multiplication

Idea: Multiplication is iterated addition.

Def. of multiplication. Let $m, n \in \mathbb{N}$, define $\alpha_m : \mathbb{N} \to \mathbb{N}$ by $\alpha_m(x) = x + m$. $(x \mapsto x+m)$

As usual, we write α_m^n for the nth iterate of α_m.

We define $m \cdot n \overset{\text{def}}{=} \alpha_m^n(0)$.

Ex. Compute $2 \cdot 3$.

$$2 \cdot 3 = \alpha_2^3(0) = \alpha_2^{\sigma(2)}(0) = \alpha_2 \circ \alpha_2^2(0)$$

$$= \alpha_2 \circ \alpha_2 \circ \alpha_2^1(0)$$

$$= \alpha_2 \circ \alpha_2 \circ \alpha_2(0)$$

$$= \alpha_2 \circ \alpha_2(0+2)$$

$$= \alpha_2 \circ \alpha_2(2)$$

$$= \alpha_2(2+2)$$

$$= \alpha_2(2 + \sigma(1))$$

$$= \alpha_2(\sigma^{\sigma(1)}(2))$$

$$= \alpha_2(\sigma \circ \sigma^1(2))$$

$$= \alpha_2(\sigma \circ \sigma(2))$$

$$= \alpha_2(\sigma(3))$$

$$= \alpha_2(4) = 4 + 2 = \sigma^2(4) = \sigma^{\sigma(1)}(4)$$

$$= \sigma \circ \sigma^1(4) = \sigma \circ \sigma(4) = \sigma(5) = 6.$$

Theorem 11. $\forall m \in \mathbb{N}$, $m \cdot 0 = 0$.

Proof. We compute $m \cdot 0 = \alpha_m^0(0)$ by def. of mult.

Now α_m^0 is the identity function by Axiom 5, so

$\alpha_m^0(0) = id(0) = 0$. Thus $m \cdot 0 = 0$. $\square$

Lemma 12. $\forall m, n \in \mathbb{N}$, $m \cdot \sigma(n) = (m \cdot n) + m$

Thm. 13. $\forall m \in \mathbb{N}$, $m \cdot 1 = m$.

Thm. 14. $\forall x, y, z \in \mathbb{N}$, $(x+y) \cdot z = xz + yz$

(Right Distributivity) Proof Sketch: Fix x, y, induct on z.

$S_{x,y} = \{z \in \mathbb{N} : (x+y) \cdot z = xz + yz\}$.

Lemma 15. If $n \in \mathbb{N}$, $0 \cdot n = 0$. : & | are both acceptable.

Lemma 16. If $n \in \mathbb{N}$, $1 \cdot n = n$

Theorem 17. (Commutativity of Multiplication)

$\forall x, y \in \mathbb{N}$, $x \cdot y = y \cdot x$.

Proof. Let $x \in \mathbb{N}$. Defining $S_x = \{y \in \mathbb{N} : xy = yx\}$,

we proceed by induction.

We have $x \cdot 0 = 0$ by Thm. 11, and $0 \cdot x = 0$ by Lemma 15.

Thus $x \cdot 0 = 0 \cdot x$, so $0 \in S_x$. (Base Case)

(induction hypothesis)

Now suppose $n \in S_x$. We compute

$$\begin{aligned} x \cdot \sigma(n) &= x \cdot n + x && \text{(by Lem. 12)} \\ &= n \cdot x + x && \text{(by hyp.)} \\ &= n \cdot x + 1 \cdot x && \text{(Lem. 16)} \\ &= (n+1) \cdot x && \text{(Thm. 14)} \\ &= \sigma(n) \cdot x && \text{(Thm. 6).} \end{aligned}$$

Thus $x \cdot \sigma(n) = \sigma(n) \cdot x$, so $\sigma(n) \in S_x$.

Since $0 \in S_x$ and $\sigma(n) \in S_x$ whenever $n \in S_x$, Axiom 4 shows $S_x = \mathbb{N}$.

That is, $S_x = \{y \in \mathbb{N} : xy = yx\} = \mathbb{N}$.

Since x was arbitrary, we conclude $xy = yx$ $\forall\, x, y \in \mathbb{N}$. $\square$